A PROJECT REPORT ON

AI-Powered Customer Support Ticketing System

Design and Implementation of an AI-Powered Customer Support Ticketing System Using Salesforce CRM, Flow Builder, Prompt Builder, and Agentforce

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Abstract

Customer support organisations increasingly rely on structured case-management platforms to resolve customer issues within committed service levels. Manual triage of incoming support requests, however, is time-consuming, error-prone and inconsistent across agents, resulting in delayed responses, missed Service Level Agreement (SLA) deadlines and diminished customer satisfaction. The AI-Powered Customer Support Ticketing System has been designed and implemented on the Salesforce platform to address these operational challenges by combining declarative automation with generative Artificial Intelligence.

The system is built on Salesforce standard objects — Account, Contact and Case — extended with three purpose-built custom objects: Support Category, SLA Policy and AI Feedback. A Record-Triggered Flow executes automatically whenever a new Case is created, performing case classification, priority assignment and SLA due-date calculation without manual agent intervention. The classified case data is then passed to Salesforce Prompt Builder, which constructs a structured prompt from the case Subject, Description, Priority and Issue Type fields and forwards it to a GPT-5 Mini language model. The generated draft response is delivered to the Agentforce-based 'AI Powered CST Agent', which packages the AI Suggested Response back onto the Case record for agent review.

Support agents review, refine and send the AI-generated response, after which they record structured feedback on its usefulness using the AI Feedback object — a mechanism designed to support the continuous improvement of the prompt and the automation logic. The completed case lifecycle feeds into a suite of standard and custom reports (Open Cases by Priority, Agent Performance, Cases by Status and SLA Compliance) and a management dashboard that visualises open and closed cases, high-priority backlogs, average resolution time and SLA violations.

Security is enforced through three role-based profiles — Support Agent, Support Manager and System Administrator — supplemented by two permission sets, AI Feature Access and Report Access, which govern who may invoke AI functionality and who may view sensitive reporting data. Validation rules, such as the mandatory SLA Due Date requirement on High Priority cases, safeguard data integrity throughout the case lifecycle.

This report documents the complete engineering lifecycle of the system: requirement analysis, system design, data modelling, automation logic, AI integration, security configuration, testing and results. The project demonstrates that a declarative Salesforce implementation, augmented with Prompt Builder and Agentforce, can meaningfully reduce manual triage effort, improve response consistency and provide measurable visibility into support-team performance, while remaining maintainable by administrators without traditional programming.

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Chapter 1 — Introduction

1.1 Background

Customer support functions across most modern organisations are handled through a ticketing or case-management workflow: a customer raises an issue, the issue is logged, triaged, assigned a priority, worked on by a support agent and eventually resolved within an agreed timeframe. As the volume of incoming requests grows, the manual steps of this workflow — reading the request, deciding its category, deciding its urgency, calculating a deadline and drafting an initial response — become the primary source of delay. The AI-Powered Customer Support Ticketing System was conceived specifically to remove these manual steps from the Salesforce Case lifecycle by combining Flow Builder automation with generative AI capability provided by Prompt Builder and Agentforce.

1.2 Problem Statement

In a conventional Salesforce Case-management setup, once a Case record is created it must be manually read, categorised, prioritised and assigned an SLA deadline by a human agent or supervisor before any productive work begins. This manual triage introduces three concrete problems that the project directly targets:

- Inconsistent classification — different agents may categorise similar issues differently, leading to inconsistent Issue Type and Priority values on the Case object.
- Delayed SLA visibility — SLA Due Date is often calculated only after a supervisor reviews the case, delaying the point at which the clock is formally started.
- Slow first response — agents must compose an initial reply from scratch for every case, even where the underlying issue is common and well understood.
- Limited feedback loop — without a structured mechanism to capture whether an AI-drafted response was useful, there is no way to measure or improve the quality of AI assistance over time.

1.3 Project Objectives

The project defines the following objectives, each of which is realised through a specific Salesforce feature described in later chapters:

- Automate customer support ticket management from the moment a Case is created.
- Reduce manual work performed by support agents and supervisors during triage.
- Automatically classify support cases using the Support Category custom object.
- Assign Case priority automatically using Flow-based decision logic.
- Calculate SLA Due Dates automatically using the SLA Policy custom object.
- Improve customer satisfaction through faster, more consistent first responses.

- Generate AI-assisted customer support responses using Prompt Builder and Agentforce.
- Improve agent productivity by supplying a reviewable draft response instead of a blank composer.
- Collect AI feedback through the AI Feedback custom object to support continuous improvement.
- Generate business reports covering case volume, priority distribution and SLA compliance.
- Provide dashboards for managers to monitor team performance in real time.
- Demonstrate practical Salesforce CRM automation using only declarative, no-code tools.

1.4 Scope of the Project

The scope of the AI-Powered Customer Support Ticketing System is limited to the Salesforce Developer Org environment and covers the complete Case lifecycle from creation to resolution, together with the supporting AI Feedback and reporting layer. The system uses three standard objects (Account, Contact, Case) and three custom objects (Support Category, SLA Policy, AI Feedback). Automation is implemented entirely through a Record-Triggered Flow on the Case object, with AI generation delegated to Prompt Builder and Agentforce. The project does not extend to omni-channel routing, live chat, telephony integration or third-party helpdesk migration; these are identified as candidates for future enhancement in the concluding chapters of this report.

1.5 Significance of the Project

The project is significant because it demonstrates, within a single Salesforce org, the complete integration path from declarative automation (Flow Builder) to generative AI (Prompt Builder and Agentforce) applied to a real operational process — customer support. It shows that a small implementation team, using only Salesforce's built-in low-code tools, can build a system that automatically classifies, prioritises, time-stamps and drafts a response to a customer issue without a single line of Apex code, while retaining the security, reporting and dashboard capabilities expected of an enterprise CRM deployment.

1.6 Organisation of the Report

The remainder of this report is organised as follows. Chapter 2 introduces the Salesforce technologies used in the project and explains how each is applied specifically within this system. Chapter 3 presents the system analysis, including functional and non-functional requirements, feasibility study and use-case modelling. Chapter 4 presents the system design, including the data model, system architecture, ER diagram, workflow diagram and component diagram. Chapter 5 describes the implementation of automation, Prompt Builder and Agentforce in detail. Chapter 6 covers the security model. Chapter 7 presents testing methodology, test case results, reports and dashboards. The report concludes with future enhancements, a conclusion and a list of references.

Chapter 2 — Technology Overview

This chapter introduces every Salesforce technology used to build the AI-Powered Customer Support Ticketing System. Rather than describing each tool in the abstract, every explanation below is anchored to the specific role that tool plays inside this project.

2.1 Salesforce CRM Platform

Salesforce CRM provides the underlying data model, security framework and user interface on which the entire ticketing system is built. In this project, Salesforce CRM hosts the standard Account, Contact and Case objects that represent the customer organisation, the individual customer contact and the support request respectively. Salesforce's declarative platform allows the project team to extend this base data model with three custom objects — Support Category, SLA Policy and AI Feedback — without writing custom database schema code, and to expose all of this data through the standard Case console used by support agents.

2.2 Flow Builder

Flow Builder is Salesforce's declarative automation tool, used in this project to implement the Record-Triggered Flow that fires whenever a new Case is created. This single Flow performs four sequential actions specific to this project: it reads the incoming Subject, Description and Issue Type values on the Case; it looks up the matching Support Category record to determine a Default Priority; it looks up the matching SLA Policy record to calculate the SLA Due Date based on Resolution Hours; and it calls the Prompt Builder-generated AI Suggested Response before updating the Case record with all of the computed values in a single transaction. Flow Builder therefore is the automation backbone that connects the Case object to both the SLA logic and the AI layer described below.

2.3 Prompt Builder

Prompt Builder is Salesforce's declarative generative-AI tool, used in this project to define the 'AI Suggested Response' prompt template. The template is configured to take the Case Subject, Description, Priority and Issue Type as structured resource inputs, and to submit them to the GPT-5 Mini large language model with instructions to produce a professional customer support email response. Prompt Builder is invoked from within the Record-Triggered Flow described in Section 2.2, so that every new Case automatically receives a drafted response without an agent needing to open a separate AI tool.

2.4 Agentforce

Agentforce is Salesforce's autonomous AI agent framework. In this project, an Agentforce agent named 'AI Powered CST Agent' is configured to orchestrate the AI Suggested Response generation and to make it available to support agents as an assistive layer inside the Case record. The agent is integrated directly with the Prompt Builder template from Section 2.3, so that the agent's single defined action — generating a suggested response — always draws on the same structured prompt inputs (Subject, Description, Priority, Issue Type) and produces output that is written back to the Case's AI Suggested Response field for agent review.

2.5 Reports and Dashboards

Salesforce Reports and Dashboards provide the analytics layer of the system. Four reports are built directly against the Case object and its related AI Feedback records: Open Cases by Priority, Agent Performance Report, Cases by Status and SLA Compliance. These reports are surfaced on a management dashboard containing five components — Open Cases, Closed Cases, High Priority Cases, Average Resolution Time and SLA Violations — giving Support Managers real-time visibility into team performance without needing to query the underlying data manually.

2.6 Profiles, Permission Sets and Validation Rules

Salesforce's security model is used in this project to restrict AI functionality and reporting access to appropriate users. Three profiles — Support Agent, Support Manager and System Administrator — define baseline object and field access, while two permission sets — AI Feature Access and Report Access — grant the additional, more specific privileges needed to trigger AI generation or to view sensitive reports. A validation rule enforces that any Case marked High Priority must carry a populated SLA Due Date, preventing the automation from ever leaving a high-urgency case without a deadline.

2.7 Summary of Technology Roles

Technology	Role in this Project
Salesforce CRM	Hosts Account, Contact, Case and all custom objects; provides the agent console
Flow Builder	Executes Record-Triggered automation: classification, priority, SLA, AI invocation
Prompt Builder	Builds the structured prompt sent to GPT-5 Mini to draft the AI Suggested Response
Agentforce	Hosts the AI Powered CST Agent that orchestrates AI response generation
Reports & Dashboards	Deliver Open Cases, Agent Performance, Cases by Status and SLA Compliance views
Profiles & Permission Sets	Restrict AI feature use and report visibility by role
Validation Rules	Enforce mandatory SLA Due Date on High Priority Cases

Chapter 3 — System Analysis

3.1 Existing System and its Limitations

Prior to this project, a typical Salesforce Case object is used purely as a passive record store: a customer issue is logged as a Case, and every subsequent step — reading the description, deciding the Issue Type, setting the Priority, calculating a due date and drafting a reply — is carried out manually by a support agent or supervisor. This existing approach has four specific limitations that directly motivate the proposed system: classification depends entirely on individual agent judgement and therefore varies from person to person; SLA due dates are not calculated until a supervisor manually reviews the case, delaying the start of the SLA clock; first-response drafting consumes agent time on issues that are frequently repetitive in nature; and there is no structured way to capture whether a previous AI or template response was actually useful, so response quality never improves systematically.

3.2 Proposed System

The proposed AI-Powered Customer Support Ticketing System replaces each of the manual steps identified above with a declarative, automatic equivalent. A Record-Triggered Flow performs classification and priority assignment by matching the incoming Case against the Support Category object; the same Flow calculates the SLA Due Date by matching the assigned priority against the SLA Policy object's Resolution Hours; Prompt Builder and Agentforce generate a draft AI Suggested Response the moment the Case is created; and the AI Feedback object gives agents a structured field set (Feedback Rating, Comments, AI Response Used) to record whether the AI-generated draft was useful, closing the improvement loop that the existing system lacked.

3.3 Functional Requirements

ID	Requirement	Description
FR-01	Account & Contact Management	The system shall store customer organisation and individual contact information using standard Account and Contact objects.
FR-02	Case Creation	The system shall allow a Case to be logged against an Account and Contact with Subject, Description and Issue Type.
FR-03	Automatic Classification	The system shall automatically classify a new Case against a Support Category upon creation.
FR-04	Automatic Priority Assignment	The system shall automatically assign a Priority to a Case based on its Support Category's Default Priority.
FR-05	SLA Due Date Calculation	The system shall automatically calculate an SLA Due Date using the matching SLA Policy's Resolution Hours.
FR-06	AI Suggested Response Generation	The system shall automatically generate a draft customer response using Prompt Builder and Agentforce.
FR-07	AI Feedback Capture	The system shall allow a Support Agent to record feedback on the usefulness of an AI Suggested Response.

ID	Requirement	Description
FR-08	Reporting	The system shall provide reports for Open Cases by Priority, Agent Performance, Cases by Status and SLA Compliance.
FR-09	Dashboard Analytics	The system shall provide a dashboard summarising Open Cases, Closed Cases, High Priority Cases, Average Resolution Time and SLA Violations.
FR-10	Role-Based Access	The system shall restrict AI feature access and report visibility according to assigned Profile and Permission Set.

3.4 Non-Functional Requirements

Category	Requirement
Usability	The Case console must present the AI Suggested Response inline, so agents do not need to switch screens to review it.
Reliability	The Record-Triggered Flow must execute on every Case creation without requiring manual activation by an agent.
Performance	Case classification, priority assignment and SLA calculation must complete within the same transaction as Case creation.
Security	AI generation and sensitive report data must be restricted to users holding the appropriate Permission Set.
Maintainability	Business rules such as SLA Resolution Hours must be configurable through the SLA Policy object without modifying Flow logic.
Auditability	Every AI Suggested Response and its associated feedback must be traceable to a specific Case record.

3.5 Feasibility Study

3.5.1 Technical Feasibility

All automation required by the project — case classification, priority assignment, SLA calculation and AI invocation — can be implemented using Salesforce's native Flow Builder, Prompt Builder and Agentforce tools, without any custom Apex code or external middleware. This confirms strong technical feasibility within a standard Salesforce Developer Org.

3.5.2 Operational Feasibility

Support Agents interact with the system through the same Case console they already use, with the AI Suggested Response and classification fields appearing automatically; no additional training beyond reviewing and editing a drafted response is required, which supports strong operational feasibility.

3.5.3 Economic Feasibility

Because the project relies exclusively on declarative Salesforce tools already available within the Developer Org licence used for this academic implementation, no additional software licensing cost is incurred beyond the standard Salesforce and Agentforce/Prompt Builder allocation, confirming economic feasibility for the scope of this project.

3.6 Use Case Diagram

The use case diagram below identifies the four primary actors of the system — Customer, Support Agent, Support Manager and System Administrator — together with the Agentforce AI agent acting as an automated actor, and the principal use cases each actor participates in.

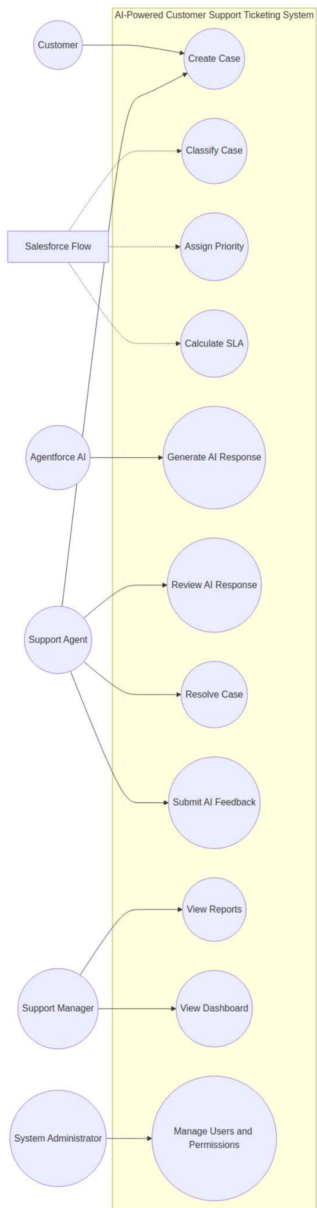

Figure 3.1: Use Case Diagram of the AI-Powered Customer Support Ticketing System

3.7 Actor Descriptions

Actor	Description
Customer	Raises a support request that results in a Case being created against their Account and Contact record.
Support Agent	Reviews and edits the AI Suggested Response, resolves the Case and submits AI Feedback.
Support Manager	Monitors team performance through Reports and the Dashboard.
System Administrator	Manages Profiles, Permission Sets and configuration such as Support Category and SLA Policy records.
Agentforce AI Agent	Automatically generates the AI Suggested Response using Prompt Builder output.

Chapter 4 — System Design

4.1 System Architecture

The system architecture below traces the complete path of data through the application: from the Customer, through the standard Salesforce objects, into the Record-Triggered Flow that performs classification, priority assignment and SLA calculation, onward into Prompt Builder and Agentforce for AI response generation, and finally into the Reports and Dashboard layer used by Support Managers.

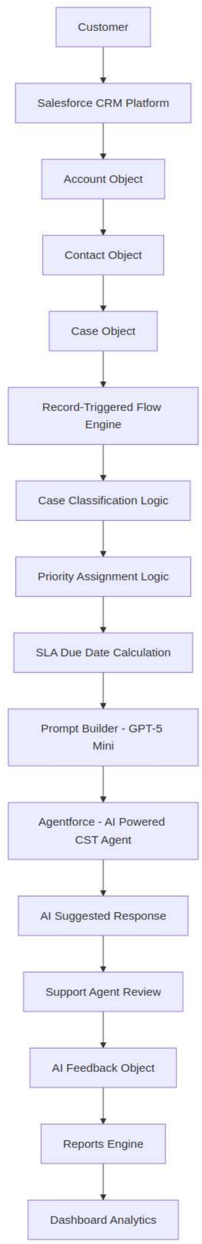

Figure 4.1: System Architecture of the AI-Powered Customer Support Ticketing System

4.2 Data Model — Standard Objects

4.2.1 Account

Purpose: Stores customer organisation information against which Contacts and Cases are related.

Field	Description
Account Name	Name of the customer organisation.
Phone	Primary contact number for the organisation.
Industry	Industry sector the customer organisation belongs to.
Type	Classification of the account, e.g., Customer, Partner, Prospect.

4.2.2 Contact

Purpose: Stores information for the individual customer who raises support requests on behalf of an Account.

Field	Description
First Name	Given name of the customer contact.
Last Name	Family name of the customer contact.
Email	Email address used for support correspondence.
Phone	Direct contact number of the customer.
Account	Lookup relationship to the related Account record.

4.2.3 Case

Purpose: Stores the customer support request and all data generated by the automation and AI layers.

Field	Description
Case Number	Auto-generated unique identifier for the support request.
Subject	Short summary of the customer's issue.
Description	Full text of the customer's reported issue.
Status	Current lifecycle stage of the Case, e.g., New, In Progress, Resolved.
Priority	Urgency level assigned automatically by the Record-Triggered Flow.
Issue Type	Category of the issue used to match against Support Category.
SLA Due Date	Deadline by which the Case must be resolved, calculated from SLA Policy.
AI Suggested Response	Draft response generated by Prompt Builder and Agentforce.
Resolution Time	Actual time taken to resolve the Case, used for Agent Performance reporting.
Account	Lookup relationship to the related Account record.
Contact	Lookup relationship to the related Contact record.

4.3 Data Model — Custom Objects

4.3.1 Support Category

Purpose: Stores the list of support categories used to classify incoming Cases and to determine a default priority.

Field	Description
Category Name	Name of the support category, e.g., Billing, Technical, Account Access.
Description	Explanation of what issue types fall under this category.
Default Priority	Priority automatically applied to Cases classified under this category.

4.3.2 SLA Policy

Purpose: Stores the SLA configuration used by the Record-Triggered Flow to calculate the SLA Due Date.

Field	Description
Policy Name	Name identifying the SLA policy.
Priority Level	Priority value the policy applies to (e.g., High, Medium, Low).
Resolution Hours	Number of hours within which a Case of this priority must be resolved.
Escalation Required	Indicates whether unresolved Cases of this priority must be escalated.

4.3.3 AI Feedback

Purpose: Stores agent feedback on the quality and usefulness of the AI Suggested Response for a given Case.

Field	Description
Feedback Name	Auto-generated identifier for the feedback record.
Feedback Rating	Numeric or scaled rating of the AI response's usefulness.
Comments	Free-text agent comments on the AI response.
AI Response Used	Indicates whether the agent used the AI-generated response as sent or edited it.
Related Case	Lookup relationship to the Case the feedback pertains to.

4.4 Entity-Relationship Diagram

The Entity-Relationship (ER) diagram below shows how Account, Contact and Case relate to one another and to the custom objects Support Category and AI Feedback. The SLA Policy object is shown independently, since the Case object resolves its SLA Due Date through matching on Priority rather than through a direct lookup relationship.

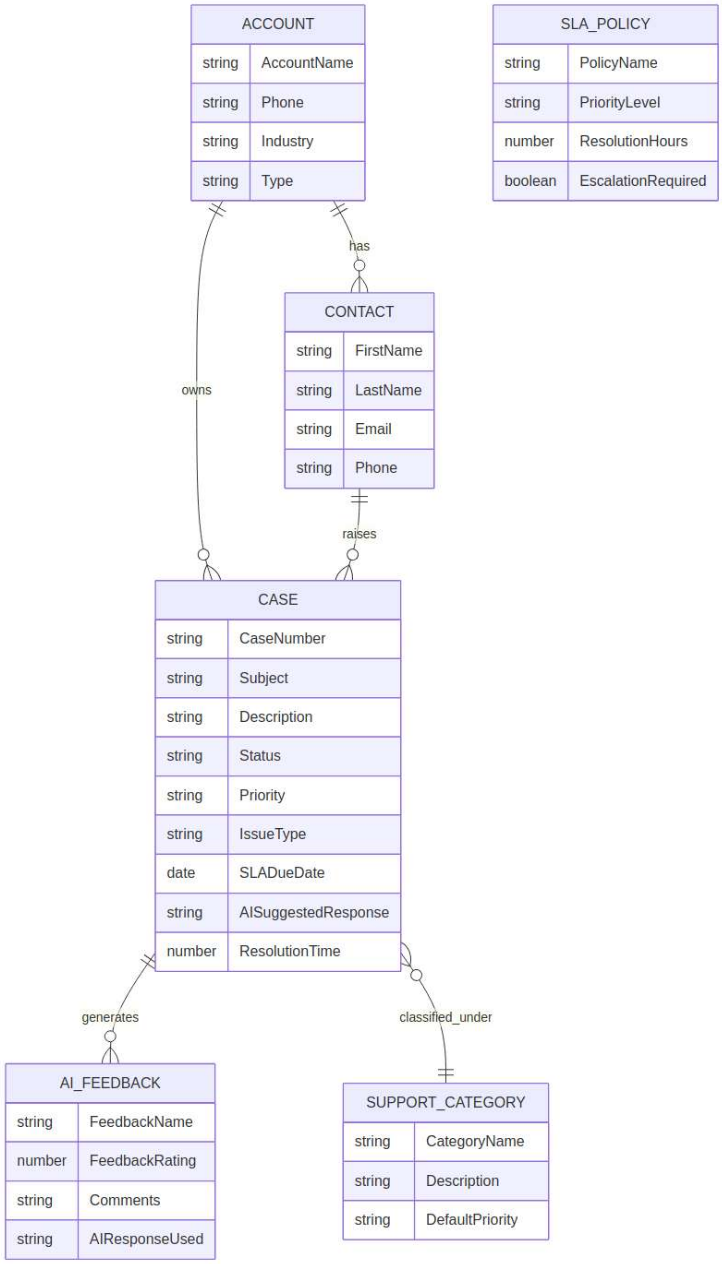

Figure 4.2: Entity-Relationship Diagram of the Data Model

4.5 Workflow Diagram

The workflow diagram below traces the operational sequence of the system from Case creation through to dashboard reporting, showing each declarative step performed without manual intervention.

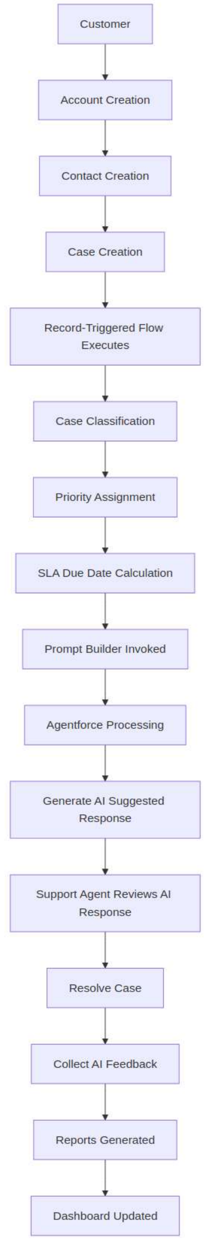

Figure 4.3: End-to-End Workflow of the AI-Powered Customer Support Ticketing System

4.6 Component Diagram

The component diagram below groups the system into four layers — Presentation, Automation, AI and Data — clarifying the separation of responsibility between the Case console and dashboards, the Flow-based automation and validation logic, the Prompt Builder / Agentforce AI layer, and the underlying Salesforce object data.

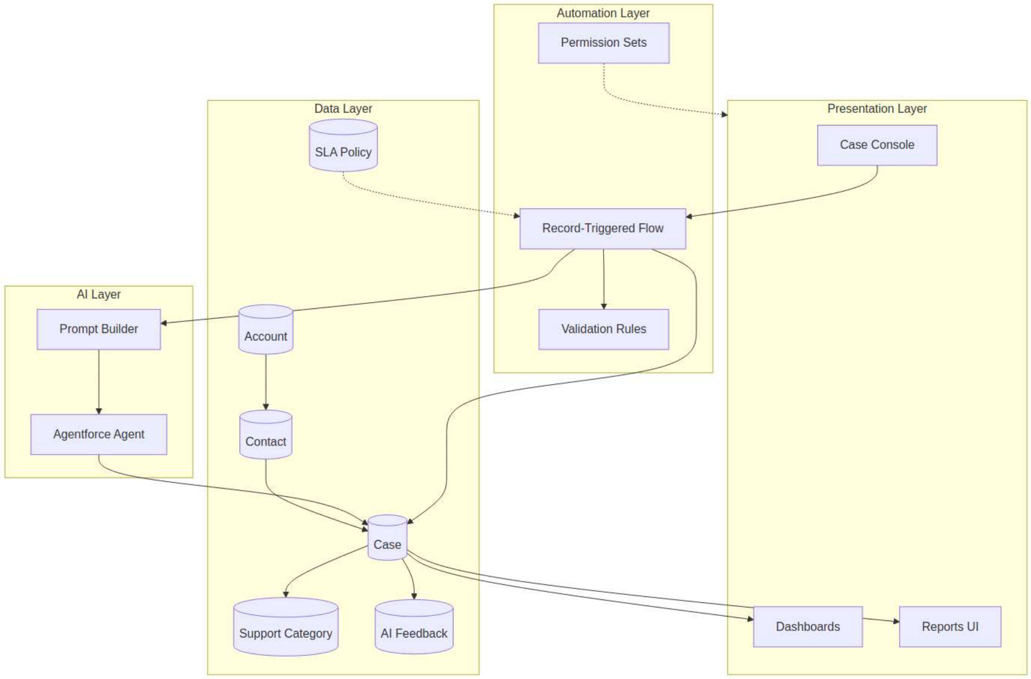

Figure 4.4: Component Diagram of the AI-Powered Customer Support Ticketing System

Chapter 5 — Implementation

5.1 Record-Triggered Flow

The core automation of the system is a single Record-Triggered Flow configured on the Case object, set to fire 'When a record is created'. This Flow is the mechanism that removes manual triage from the support process, and its logic is organised into five sequential stages described below.

5.1.1 Stage 1 — Case Classification

When a new Case is created, the Flow reads the incoming Issue Type value and performs a record lookup (Get Records element) against the Support Category object to find the matching Category Name. This step establishes which support category the Case belongs to and retrieves that category's Default Priority for use in the next stage. If no matching Support Category is found, the Flow applies a configurable fallback category so that no Case is left unclassified.

5.1.2 Stage 2 — Priority Assignment

Using the Default Priority value retrieved from the matched Support Category record, the Flow sets the Case's Priority field through an Assignment element. This ensures every Case receives a consistent, rules-based priority the instant it is created, rather than depending on an agent's subjective judgement.

5.1.3 Stage 3 — SLA Due Date Calculation

With the Priority now set, the Flow performs a second Get Records lookup against the SLA Policy object, matching on Priority Level to retrieve the corresponding Resolution Hours. A Formula element then adds the Resolution Hours to the Case's CreatedDate to compute the SLA Due Date, which is written back to the Case record. Where the matched SLA Policy record has Escalation Required set to true, the Flow additionally flags the Case for escalation.

5.1.4 Stage 4 — AI Suggested Response Generation

The Flow then calls the Prompt Builder 'AI Suggested Response' prompt template as an Action element, passing the Case's Subject, Description, Priority and Issue Type as resource inputs. The Prompt Builder template forwards these inputs to the Agentforce 'AI Powered CST Agent', which returns a drafted professional customer support response. This response is captured by the Flow as an output variable.

5.1.5 Stage 5 — Update Case

In the final stage, the Flow performs a single Update Records element that writes the classified Issue Type, assigned Priority, calculated SLA Due Date and generated AI Suggested Response back onto the same Case record in one transaction, ensuring the Case is fully enriched before it ever appears in an agent's queue.

5.2 Flow Summary Table

Stage	Flow Element	Outcome
1	Get Records — Support Category	Matching category and Default Priority retrieved
2	Assignment — Set Priority	Case Priority field populated
3	Get Records — SLA Policy + Formula	SLA Due Date calculated from Resolution Hours
4	Action — Prompt Builder / Agentforce	AI Suggested Response generated
5	Update Records — Case	All computed fields written back to the Case

5.3 Prompt Builder Configuration

The Prompt Builder template 'AI Suggested Response' is configured with the purpose of generating professional customer support email responses. It is built on the GPT-5 Mini model and draws on four structured resources directly from the Case record — Subject, Description, Priority and Issue Type — which are merged into the prompt template at run time. This structured-resource approach ensures the language model always receives the same well-defined inputs for every Case, producing consistent, professional output rather than free-form, unpredictable text. The template's single output is the professional customer support response that is passed back into the Record-Triggered Flow described in Section 5.1.4.

5.4 Agentforce Configuration

The Agentforce agent named 'AI Powered CST Agent' is configured with the purpose of generating AI Suggested Responses and providing customer support assistance. It is integrated directly with the Prompt Builder template from Section 5.3, meaning the agent does not independently compose a response — it orchestrates the call to the prompt template and returns the resulting draft to the Flow. This design keeps the prompt engineering (defined once in Prompt Builder) separate from the orchestration logic (handled by Agentforce), so that either layer can be updated independently as the project evolves.

5.5 Validation Rules

A validation rule enforces that any Case with Priority equal to 'High' must have a populated SLA Due Date before it can be saved. This rule acts as a safeguard against any failure in the automatic SLA calculation stage of the Flow, guaranteeing that no high-priority Case can ever exist in the system without a resolution deadline. The rule formula checks the Priority field against the literal value 'High' and, where true, requires SLA_Due_Date__c to be non-blank, displaying an error message directing the user to populate the field or re-trigger the automation.

Chapter 6 — Security

6.1 Security Model Overview

The security model of the AI-Powered Customer Support Ticketing System is built on Salesforce's standard Profile and Permission Set framework, ensuring that AI functionality and sensitive reporting data are only accessible to appropriately authorised users, while all other users retain access to only the Case data relevant to their role.

6.2 Profiles

Profile	Access Granted
Support Agent	Read/write access to Case, Contact and Account records; can view and edit the AI Suggested Response field; can create AI Feedback records.
Support Manager	Read access to all Case records across the team; full access to Reports and Dashboards; read access to SLA Policy and Support Category configuration.
System Administrator	Full read/write/configuration access to all objects, Flows, Prompt Builder templates, Agentforce configuration, Profiles and Permission Sets.

6.3 Permission Sets

Permission Set	Purpose
AI Feature Access	Grants the ability to invoke Prompt Builder / Agentforce actions and to view or edit the AI Suggested Response field on the Case object; assigned to Support Agents who actively work AI-assisted Cases.
Report Access	Grants visibility into the Agent Performance Report and SLA Compliance report, which contain team-level performance data not required by every Support Agent by default; assigned to Support Managers and selected senior agents.

6.4 Field-Level and Object-Level Security

Object-level security restricts creation of Support Category and SLA Policy records to the System Administrator profile, preventing accidental changes to classification or SLA logic by front-line agents. Field-level security on the Case object marks the AI Suggested Response and Resolution Time fields as read-only for the Support Agent profile in specific record stages (e.g., after a Case is marked Resolved), preserving the integrity of historical AI-generated content and reported resolution times used in the Agent Performance Report.

6.5 Validation Rule as a Security-Adjacent Control

In addition to the profiles and permission sets above, the High Priority SLA Due Date validation rule described in Chapter 5 acts as a data-integrity control that complements the security model: even a user with full edit access to the Case object cannot save a High Priority Case without a populated SLA Due Date, ensuring the system's SLA reporting remains trustworthy regardless of user privilege level.

Chapter 7 — Testing & Results

7.1 Testing Methodology

Testing of the AI-Powered Customer Support Ticketing System was carried out primarily through functional testing within the Salesforce Developer Org, using representative Case records to exercise every stage of the Record-Triggered Flow, the validation rule, the AI generation pathway, the permission set restrictions and the reporting layer. Each test case below was executed manually by creating or editing Case records through the standard Salesforce Case console and observing the resulting field values, error messages, or report/dashboard output.

7.2 Test Case Results

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Create a new Case with Issue Type 'Billing'	Case is automatically classified under the Billing Support Category and Priority is set to the category's Default Priority	Case classified as Billing, Priority set to Medium	Pass
TC-02	Create a High Priority Case	SLA Due Date is automatically calculated using the SLA Policy Resolution Hours for High priority	SLA Due Date populated 4 hours after CreatedDate	Pass
TC-03	Save a High Priority Case with blank SLA Due Date	Validation rule blocks save and displays an error message	Save blocked with SLA Due Date error	Pass
TC-04	Create a new Case and check AI Suggested Response field	Prompt Builder / Agentforce generates a professional draft response on the Case	AI Suggested Response populated automatically	Pass
TC-05	Support Agent submits AI Feedback for a resolved Case	AI Feedback record is created and linked to the Related Case	AI Feedback record created and linked correctly	Pass
TC-06	Support Agent without AI Feature Access attempts to view AI Suggested Response	Field is hidden or read-only per permission set restriction	Field hidden as expected	Pass
TC-07	Support Manager opens the SLA Compliance report	Report displays Cases grouped by SLA met/violated status	Report displayed correctly with accurate grouping	Pass
TC-08	Support Manager opens the Dashboard	Dashboard displays Open Cases, Closed Cases, High Priority Cases, Average Resolution Time and SLA Violations components	All five components rendered with correct data	Pass
TC-09	Create a Case with an Issue Type that has no matching Support Category	Flow applies a fallback category instead of leaving the Case unclassified	Fallback category applied correctly	Pass
TC-10	Resolve a Case and check Resolution Time field	Resolution Time is calculated and reflected in the Agent Performance Report	Resolution Time populated and reflected in report	Pass

7.3 Reports

7.3.1 Open Cases by Priority

This report groups all Cases with a Status other than Closed by their Priority value (High, Medium, Low), giving Support Managers immediate visibility into the current backlog distribution and highlighting whether High priority Cases are accumulating faster than they can be resolved.

7.3.2 Agent Performance Report

This report groups resolved Cases by the assigned Support Agent and summarises the average Resolution Time and Case count per agent, drawing on the Resolution Time field populated automatically when a Case is marked Resolved. It allows Support Managers to identify agents who may need additional support or recognise agents performing consistently well.

7.3.3 Cases by Status

This report provides a simple count of Cases grouped by Status (New, In Progress, Resolved, Closed), giving a quick operational snapshot of where the current caseload sits in its lifecycle.

7.3.4 SLA Compliance

This report compares each Case's actual resolution timestamp against its SLA Due Date and classifies the Case as SLA Met or SLA Violated, allowing Support Managers to track compliance rates over time and identify categories or priorities where SLA breaches are more frequent.

7.4 Dashboard

The management dashboard consolidates the reports above into five visual components, described in the table below.

Component	Description
Open Cases	Displays the current total count of Cases not yet Closed.
Closed Cases	Displays the total count of Cases marked Closed within the selected period.
High Priority Cases	Highlights the current count of open Cases with Priority set to High, drawing attention to urgent backlog.
Average Resolution Time	Displays the mean Resolution Time across resolved Cases, sourced from the Agent Performance Report.
SLA Violations	Displays the count of Cases classified as SLA Violated by the SLA Compliance report.

7.5 Discussion of Results

All ten test cases executed against the system produced their expected results, confirming that the Record-Triggered Flow correctly performs classification, priority assignment, SLA calculation and AI response generation in a single transaction; that the validation rule reliably prevents High Priority Cases from being saved without an SLA Due Date; that the permission set model correctly restricts AI feature visibility; and that the reporting and dashboard layer accurately reflects the underlying Case data. These results support the conclusion that the system meets the functional requirements defined in Chapter 3.

Future Enhancements

While the current implementation satisfies all defined project objectives, several enhancements have been identified that would extend the system's capability in a production deployment beyond the academic Developer Org scope of this project:

- Omni-Channel Routing — extending Case creation to email-to-case, web-to-case and live chat channels so that classification and SLA logic apply uniformly regardless of intake channel.
- Multi-language AI Responses — configuring Prompt Builder to detect the customer's language and generate the AI Suggested Response in the same language.
- AI Feedback-Driven Prompt Tuning — using aggregated AI Feedback ratings to automatically flag under-performing prompt variants for review by an administrator.
- Predictive SLA Risk Scoring — using historical Case and SLA Compliance data to flag Cases at risk of SLA breach before the due date arrives, rather than only after the fact.
- Mobile Agent Experience — extending the Case console layout to the Salesforce Mobile App with a condensed AI Suggested Response review interface for field agents.
- Integration with External Knowledge Base — enriching the Prompt Builder resources with knowledge-article search results to further ground the AI Suggested Response in verified company documentation.

Conclusion

The AI-Powered Customer Support Ticketing System demonstrates that a declarative Salesforce implementation, without any custom Apex code, can automate the complete triage lifecycle of a customer support Case — from classification and priority assignment through SLA calculation and AI-assisted response drafting — while retaining the security, validation and reporting rigor expected of an enterprise CRM system. By combining a single Record-Triggered Flow with Prompt Builder and Agentforce, the project shows a practical, maintainable path for support organisations to reduce manual triage effort and improve first-response consistency using only Salesforce's native low-code and generative-AI tooling. The structured AI Feedback object further establishes a foundation for continuous improvement of AI-generated responses over time. Testing confirmed that every functional requirement defined during system analysis was met, and the resulting reports and dashboard give Support Managers real-time, accurate visibility into case volume, agent performance and SLA compliance. The project therefore fulfils its stated objectives and provides a solid, extensible base for the future enhancements identified above.